

Chit 20 — MongoDB: Indexing + Aggregation (Student Marks)

Indexing Operations

```
use lab_db

// 1. Create Collection
db.createCollection('Student')

// 2. Insert Documents
db.Student.insertMany([
  { Rollno:1, name:'Navin', subject:'DMSA', marks:78 },
  { Rollno:2, name:'anusha', subject:'OSD', marks:75 },
  { Rollno:3, name:'ravi', subject:'TOC', marks:69 },
  { Rollno:4, name:'veena', subject:'TOC', marks:70 },
  { Rollno:5, name:'Pravini',subject:'OSD', marks:80 },
  { Rollno:6, name:'Reena', subject:'DMSA', marks:50 },
  { Rollno:7, name:'Geeta', subject:'CN', marks:90 },
  { Rollno:8, name:'Akash', subject:'CN', marks:85 }
])

// 3. Create Single Index
db.Student.createIndex({ Rollno: 1 })

// 4. Create Compound Index
db.Student.createIndex({ subject: 1, marks: -1 })

// 5. Create Unique Index
db.Student.createIndex({ Rollno: 1 }, { unique: true })

// 6. Show Index Information
db.Student.getIndexes()

// 7. Remove Index
db.Student.dropIndex({ subject: 1, marks: -1 })
```

Aggregation Queries

2. Max marks of each subject

```
db.Student.aggregate([
  { $group: { _id: '$subject', Max_Marks: { $max: '$marks' } } }
])
```

3. Min marks of each subject

```
db.Student.aggregate([
  { $group: { _id: '$subject', Min_Marks: { $min: '$marks' } } }
])
```

4. Sum of marks of each subject

```
db.Student.aggregate([
  { $group: { _id: '$subject', Total_Marks: { $sum: '$marks' } } }
])
```

5. Average marks of each subject

```
db.Student.aggregate([
  { $group: { _id: '$subject', Avg_Marks: { $avg: '$marks' } } }
])
```

6. First record of each subject

```
db.Student.aggregate([
  { $sort: { Rollno: 1 } },
  { $group: { _id: '$subject', First_Student: { $first: '$name' },
    First_Marks: { $first: '$marks' } } }
])
```

7. Last record of each subject

```
db.Student.aggregate([
  { $sort: { Rollno: 1 } },
  { $group: { _id: '$subject', Last_Student: { $last: '$name' },
    Last_Marks: { $last: '$marks' } } }
])
```

8 & 9. Count number of records of each subject

```
db.Student.aggregate([
  { $group: { _id: '$subject', Count: { $sum: 1 } } }
])
```